

Planetary Defence Events as Technosignatures

What Would Be Detectable, and What Would Not

Aaron Garcia

aaron@garcia.ltd

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Summary

Every civilisation that lives on a planet will eventually be in the path of something large. Over ten thousand years, a Tunguska-class impact is close to certain; over a million, something far worse. A civilisation that intends to persist must therefore either move the object or move itself. Neither is quiet.

This paper asks whether that activity would be visible from another star, and answers honestly: mostly not, and not in the way the idea first suggests. An earlier draft of this work claimed that moon-scale deflection events would reach naked-eye brightness at a hundred light-years. That claim was wrong by roughly three orders of magnitude, and correcting it changes the paper. What survives is narrower and, I think, more useful: the optical flash is a dead end, the infrared debris signature is real and already detected around other stars for natural collisions, and the technosignature question is therefore not “can we see the event” but “can we distinguish a managed event from an unmanaged one”.

1. Why the Activity Should Be Common

Impact frequency follows a power-law size distribution with reasonably well-characterised parameters (Harris and D’Abramo, 2015). Over a ten-thousand-year window — short by civilisational standards — a 50–60 metre impactor is near-certain, a 100–200 metre event is roughly even odds, and a kilometre-class object is a percent-level risk. Extend the window and the tail becomes the story.

The argument for treating defence as a widespread activity is that it is not optional and not culturally contingent. A civilisation may or may not build radio transmitters, Dyson spheres, or interstellar probes; those are choices. Being hit is not a choice. Any civilisation that survives long enough to be worth detecting has, with high probability, either developed deflection capability or been fortunate. This makes defensive activity a more universal expectation than communication — which is the entire reason it is worth asking what it looks like from outside.

2. The Optical Transient Does Not Work

It is tempting to imagine a deflection or a prevented collision as a visible flash. The arithmetic forbids it, and the arithmetic is simple enough to state completely.

At 100 light-years — 30.7 parsecs — the distance modulus is 2.43 magnitudes. For an event to appear at apparent magnitude -2 , it must have absolute magnitude -4.4 , which is 9.3 magnitudes brighter than the Sun: about 5,000 solar luminosities, or 1.9×10^{30} watts. Sustained for five hours that is 3.5×10^{34} joules — thousands of times the total energy of the Moon-forming impact (itself uncertain to an order of magnitude, and the most violent event in this solar system’s history). A deflection changes an object’s velocity; it does not vaporise a moon, and even vaporising a moon falls orders of magnitude short.

The error in the earlier draft was a category confusion between an impact flash and a stellar event. It is worth stating plainly because the same confusion is easy to make in reverse: the reason giant impacts are detectable at all is not that they are bright, but that their debris stays warm for a long time.

3. What Is Actually Detectable: Warm Debris

What persists after a planetary-scale collision is not the flash but the dust. The impact grinds and vaporises material into fragments whose collective surface area is enormous compared with the bodies that produced them, and that surface intercepts the star's own light and re-emits it in the infrared — for millennia, powered by the star rather than by any stored heat from the impact, which dissipates far sooner. The observed infrared excesses of impact-debris systems sit near a tenth of a percent of the stellar output. (As a labelled illustration only: even radiating the Moon-forming impact's own energy, of order 10^{31} joules, spread across a year would give a similar fraction — but that is not the sustaining mechanism; the cross-section is.)

This is not speculation. Warm dust interpreted as the debris of recent planetary-scale collisions has been detected around several stars. HD 172555 shows an extreme mid-infrared silica feature whose glassy, vapour-condensed mineralogy points to a hypervelocity planetary-scale impact (Johnson et al., 2012; Lisse et al., 2009), and has since been resolved with ALMA and observed with JWST. BD +20 307 carries warm dust at roughly 420 K within an astronomical unit of the star, most plausibly from a catastrophic collision between rocky bodies (Weinberger et al., 2011). Giant impacts around other stars are, in other words, already an observed phenomenon.

That reframes the problem entirely. We do not need a new detection method for planetary-scale events. We need a way to tell whether an event was managed.

4. The Real Question: Managed Versus Unmanaged

A natural collision and a deflection campaign both disturb a planetary system, but they should not disturb it identically. Four discriminants seem worth pursuing, and their difficulty varies enormously.

4.1 Isotopic Anomaly

Nuclear deflection would inject fission products into circumstellar debris. Technetium-99 is the most-cited candidate, and requires a correction that the earlier draft got wrong: technetium is produced naturally, by slow neutron capture in thermally pulsing asymptotic giant branch stars — its detection in red giants by Merrill in 1952 was the first direct proof of stellar nucleosynthesis. The technosignature argument therefore has to be stated more carefully. Technetium-99 has a half-life near 210,000 years and no local production route in a cold main-sequence planetary system; its presence in the debris of such a system is anomalous precisely because it should long since have decayed and cannot be replenished there. Caesium-137 and strontium-90, with half-lives near thirty years, would indicate activity within living memory. Whether any of these are measurable in an exoplanetary debris spectrum at interstellar distance is, at

present, doubtful — they are listed as the discriminant that would be decisive if the measurement ever becomes possible, not as one that is available now.

4.2 Statistical Absence

A civilisation that clears hazardous objects should, over time, produce a debris population depleted in exactly the orbits that threaten its planet. This is an argument from absence, and absence is weak evidence — but it is statistical rather than anecdotal, and would apply across a survey of systems rather than to any single one. It also requires a level of small-body census in other systems that current instruments cannot deliver.

4.3 Timing

Natural collisions arrive at random. A deflection campaign against a known threat would show energy release repeatedly, at intervals matched to an orbital period, on an object whose orbit is being progressively altered. Repetition with orbital periodicity is the least exotic of the four discriminants and the one most likely to be caught by long-baseline monitoring of a system already known to host warm dust.

4.4 Thermal Signature of Evacuation

Moving a population off-world is a sustained energetic activity rather than a single event, and would appear as prolonged infrared emission with a spectral character unlike collision debris. This is the most speculative of the four and depends on assumptions about propulsion that cannot be constrained from here.

5. Honest Detection Limits

The following table replaces the detection thresholds of the earlier draft, which were derived from the erroneous optical estimate.

Signature	Physical basis	Detectable at ~100 ly?
Optical transient from deflection	Requires ~5,000 L _☉ ; deflection energies fall orders of magnitude short	No — ruled out by the arithmetic in Section 2
Infrared excess from collision debris	Warm dust radiating over years; observed at HD 172555, BD +20 307	Yes — already achieved for natural events
Repetition with orbital periodicity	Multiple energy releases on one body over an orbital period	Plausible with long-baseline monitoring of known warm-dust systems
Isotopic anomaly (Tc-99, Cs-137)	Fission products absent from cold main-sequence systems	Not with current or near-term instruments
Statistical orbital clearing	Depletion of threatening orbits across a survey	Requires small-body censuses beyond present capability

Read as a whole, the table says something unglamorous: the events are detectable, the intent behind them is not — at least not yet, and not at a hundred light-years. Anyone hoping for a decisive planetary-defence technosignature in the current

instrument generation should read that as a discouraging result, and it is offered as one.

6. What This Means for the Search

If the discriminants are mostly out of reach, the useful conclusion is not that the category should be abandoned but that it should be attached to observations already being made. Systems with warm dust from suspected giant impacts are a small, named, actively studied population. Monitoring them for repetition on orbital timescales costs little beyond what is already spent, and the null result is informative: it constrains how common managed deflection is among nearby systems, or at least how common it is in a form we could notice.

There is also a symmetry worth stating. The same reasoning that makes defensive activity a plausible technosignature makes our own future activity one. A civilisation that begins deflecting objects in earnest becomes, in principle, visible to anyone applying this logic in reverse. That is not an argument for or against doing it; it is a reminder that the observable is two-way.

7. Limitations

This is a framing paper by an independent researcher, not an observational proposal, and it carries the limitations that implies. No new radiative transfer modelling has been done; the energy estimates in Sections 2 and 3 are order-of-magnitude arithmetic that any astronomer can and should check. The behavioural premise — that civilisations facing impacts will act, and act in ways involving large energy release — is not testable and never will be. The isotopic discriminant depends on measurement capabilities that do not exist. And the four discriminants of Section 4 have not been modelled quantitatively against the natural-collision baseline, which is the obvious next piece of work and requires someone with the tools to do it properly.

Prior work touching this territory should also be acknowledged, which the earlier draft did not do: planetary defence has been discussed alongside space debris and SETI in the geoengineering-technosignature literature, and solar-system searches for artefacts have been proposed as adjacent to planetary defence programmes. What appears to be missing is a systematic treatment of defence events as a signature class with stated detection limits, which is what this paper attempts.

8. Open Invitation

The specific help this needs is quantitative: someone able to model the debris signature of a deflection campaign against the natural giant-impact baseline, and say whether the difference survives at interstellar distance. If it does not, that is worth publishing too. Criticism of the arithmetic in Sections 2 and 3 is particularly welcome — it is deliberately shown in full so that it can be checked.

Contact: aaron@garcia.ltd

Appendix A: Document Metadata

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Appendix B: Glossary

Term	Definition
AGB star	Asymptotic giant branch — an evolved star; site of slow-neutron-capture nucleosynthesis
Absolute magnitude	Apparent brightness an object would have at 10 parsecs
Debris disk	Circumstellar dust produced by collisions between small bodies
Distance modulus	The difference between apparent and absolute magnitude, set by distance
Infrared excess	Emission above the stellar photosphere, indicating circumstellar dust
Parsec (pc)	3.26 light-years
s-process	Slow neutron capture; produces technetium in AGB stars
Technosignature	An observable indicating technological activity

Appendix C: Methodology and Transparency Statement

Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The analytical framework reflects professional history, focusing on practical application. The author is not an astronomer; the arithmetic in this paper is shown in full precisely so that astronomers can check it.

Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools — specifically Claude, Google Gemini, and ChatGPT — to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial

accessibility. As a writer with dyslexia, I leverage these models as assistive technology to refine grammar and streamline sentence structure.

Verification and Integrity

An earlier version of this paper contained three physics errors: an optical-brightness estimate wrong by roughly three orders of magnitude, an incorrect claim that technetium-99 has no natural production pathway, and a mistaken tritium supply-chain constraint that ignores in-situ breeding from lithium-6 deuteride in staged thermonuclear devices. All three have been corrected or removed in this version, and the corrections are described in the text rather than buried. No external peer review has been sought or obtained for this work; no reviewers, advisors, or institutions have been consulted.

Appendix D: References

Citation	Reference
Cheng et al., 2023	“Momentum transfer from the DART mission kinetic impact on asteroid Dimorphos.” <i>Nature</i> 616, 457–460.
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